Manual for Fermi-GBM Spectral Analysis using XSPEC

GBM Analysis Log

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1 Introduction

This manual documents the standard procedure for spectral analysis of Gamma-Ray Bursts (GRBs) using Fermi-GBM data. The workflow utilizes `gtburst` for data preparation and `XSPEC` for spectral fitting. The analysis focuses on the Band Function (GRBM), error estimation, and flux calculation.

2 Data Loading

The analysis begins by loading the spectra, background files, and response matrices generated by `gtburst`.

1. Open the terminal in the `current/` directory containing the exported files.
2. Launch XSPEC and run the master script:

```
XSPEC12> @loadData_bn250919020_int01.xcm
```

This command automatically loads the source spectra (`.pha`), background models (`.bak`), and response matrices (`.rsp`) for all detectors (typically NaI 6, 7, 8 and BGO 1).

3 Data Preparation and Cleaning

Before fitting, the statistical method must be defined and "bad" data channels (electronic noise) must be removed.

3.1 Statistic Selection

Since `gtburst` exports **Total Counts** (Source + Background), we use the Poisson-based **PG-Stat** statistic, which is superior to Chi-Squared for low-count regimes.

```
XSPEC12> statistic pgstat
```

3.2 Ignoring Bad Channels

GBM detectors have systematic noise at the spectral edges (e.g., the Iodine K-edge at 33 keV and electronic rollover). We mask these regions:

```
! Ignore low energy noise (< 40 keV) in NaI detectors (Groups 1-3)
XSPEC12> ignore 1-3: **-40.0 900.0-**

! Ignore noise (< 300 keV) in BGO detector (Group 4) lower range 200kev
XSPEC12> ignore 4: **-250.0 40000.0-**
```

4 Model Definition

We fit the data with the empirical **Band Function** (GRBM), multiplied by a constant factor to account for cross-calibration uncertainties between detectors.

```
XSPEC12> model const*grbm
```

4.1 Parameter Initialization

- **factor (const):** Fix to 1.0 for the first detector (Reference). Allow it to float for others.
- **alpha:** Low-energy index (typical start: -1.0).
- **beta:** High-energy index (typical start: -2.5).
- **tem:** Characteristic energy (typical start: 200 keV).
- **norm:** 0.02

```
! Example: Freeing the constant factor for detectors 2, 3, and 4
XSPEC12> untie 6 11 16
XSPEC12> thaw 6 11 16
```

5 Fitting

Run the fit algorithm to minimize the statistic.

```
XSPEC12> fit 100
```

Check the **Test Statistic** (Reduced PG-Stat). A value near 1.0 indicates a good fit.

6 Plotting (Unfolded Spectrum)

To visualize the physical spectrum ($E^2F(E)$ or νF_ν) and residuals.

6.1 Plot Setup

```
XSPEC12> cpd /xw        ! Set plot device to screen
XSPEC12> setplot energy ! Plot in Energy space (keV)
XSPEC12> plot eeuf resid ! Plot Unfolded Spectrum + Residuals
```

6.2 Formatting in PLT

Enter interactive plotting mode (`iplot`) to apply log scales and save the image.

```
XSPEC12> iplot
PLT> log x on       ! Logarithmic X-axis
PLT> log y on       ! Logarithmic Y-axis
PLT> rescale        ! Refresh view
PLT> cpd spectrum.png/PNG  ! Set output to PNG file
PLT> plot           ! Save the file
PLT> exit
```

7 Calculations: Flux, Fluence, and E_{peak}

7.1 Flux Calculation

Calculate the flux in the standard 10–1000 keV band using the reference detector (Group 1).

```
XSPEC12> flux 10.0 1000.0 err 100
```

Report the "Energy Flux" (ergs/cm^2/s) for Spectrum 1.

7.2 Fluence Calculation

Fluence is the total energy received over the burst duration (T_{dur}).

$$\text{Fluence (erg cm}^{-2}) = \text{Flux} \times T_{dur} \tag{1}$$

7.3 Peak Energy (E_{peak})

The model parameter `tem` is the characteristic energy. E_{peak} is derived as:

$$E_{peak} = E_{tem} \times (2 + \alpha) \tag{2}$$

8 Error Analysis

To determine the 90% confidence intervals for the parameters.

```
! Syntax: error <delta_chi> <param_indices>
! 2.706 corresponds to 90% confidence
XSPEC12> error 2.706 2 3 4
```

8.1 Error Propagation for E_{peak}

Since E_{peak} depends on both α and tem, we approximate the error using the fractional error of tem (dominant term):

$$\Delta E_{peak} \approx E_{peak} \times \left(\frac{\Delta tem}{tem} \right) \tag{3}$$

Note: For asymmetric errors, calculate the upper and lower bounds separately by substituting the upper/lower limits of tem into the E_{peak} equation.

9 Final Results Table

Parameter	**Best Fit Value**	**90% Confidence Interval**
Alpha (α)	-0.72	$[-0.74, -0.69]$
Beta (β)	-2.77	$[-2.94, -2.66]$
E_{tem} (keV)	392.3	$[355.2, 406.2]$
E_{peak} (keV)	502.5	$[446.1, 532.5]$
Flux (10-1000 keV)	4.41×10^{-6}	$\pm 0.07 \times 10^{-6}$ erg/cm^2/s
Fluence	2.20×10^{-5}	erg/cm^2

Table 1: Spectral analysis results for GRB 250919020.